[INSTITUTION NAME]

(An Autonomous Institution affiliated to [University Name])

COURSE AND SYLLABUS INFORMATION

Curriculum Structure and Detailed Syllabi

Sample: B.E. / B.Tech. Programme | Regulations 2026

Template document. Replace bracketed items and adjust values to match your institution's actual policies and the rules of the affiliating university or regulatory body.

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1. PROGRAMME OVERVIEW

1.1 **Programme:** B.E. / B.Tech. in [Computer Science and Engineering] (illustrative). Duration: 4 years / 8 semesters. Total credits: [160].

1.2 **Programme Educational Objectives (PEOs):**

(a) Graduates will apply sound engineering knowledge to analyse and solve real-world problems.

(b) Graduates will progress in their professional careers or pursue higher studies and research.

(c) Graduates will demonstrate ethics, teamwork, communication, and a commitment to lifelong learning.

1.3 The curriculum is framed by the Board of Studies and approved by the Academic Council, with periodic revision based on industry and alumni feedback.

2. COURSE CATEGORIES AND CREDIT DISTRIBUTION

2.1 The curriculum comprises the following categories (illustrative):

Category	Code	Credits
Humanities and Social Sciences	HS	[10]
Basic Sciences (Mathematics, Physics, Chemistry)	BS	[25]
Engineering Sciences	ES	[22]
Professional Core	PC	[50]
Professional Electives	PE	[15]
Open Electives	OE	[12]
Project Work, Internship, Seminar	PW	[20]
Mandatory Courses (non-credit)	MC	-
Total		**[154 - 160]**

3. SEMESTER-WISE COURSE STRUCTURE: YEAR I

Semester I

Code	Course Title	Cat.	L-T-P	Credits
[MA101]	Calculus and Linear Algebra	BS	3-1-0	4
[PH101]	Engineering Physics	BS	3-0-2	4
[CS101]	Problem Solving and Programming in Python	ES	3-0-2	4
[EN101]	Technical English	HS	2-0-2	3
[ME101]	Engineering Graphics	ES	2-0-2	3
	Total			**18**

Semester II

Code	Course Title	Cat.	L-T-P	Credits
[MA102]	Differential Equations and Transforms	BS	3-1-0	4
[CH101]	Engineering Chemistry	BS	3-0-2	4
[EE101]	Basic Electrical and Electronics Engineering	ES	3-0-2	4
[CS102]	Data Structures	PC	3-0-2	4
[HS102]	Universal Human Values	HS	2-0-0	2
	Total			**18**

4. SEMESTER-WISE COURSE STRUCTURE: YEARS II TO IV (OUTLINE)

4.1 Year II (Semesters III and IV): Discrete Mathematics, Object Oriented Programming, Digital Logic and Computer Organisation, Database Management Systems, Operating Systems, Design and Analysis of Algorithms, Probability and Statistics, and laboratory courses.

4.2 Year III (Semesters V and VI): Computer Networks, Theory of Computation, Software Engineering, Compiler Design, Machine Learning, Web Technologies, Professional Electives I-II, Open Elective I, Mini Project, and Internship.

4.3 Year IV (Semesters VII and VIII): Cloud Computing, Cryptography and Network Security, Professional Electives III-V, Open Elective II, Project Work (Phase I and II), and Seminar.

4.4 The complete course-wise structure with codes and credits is available in the department's approved curriculum booklet.

5. SAMPLE DETAILED SYLLABI

5.1 The following sample syllabi show the standard format used across the institution. Each course lists objectives, unit-wise topics with contact hours, outcomes, and reference books.

[CS101] - Problem Solving and Programming in Python

L-T-P-C: 3-0-2-4

Course Objectives

(a) To introduce computational thinking and algorithmic problem solving.

(b) To develop programs using Python's data types, control structures, and functions.

(c) To use files, modules, and libraries for practical applications.

Syllabus

Unit	Topics	Hours
I	Computational thinking, algorithms, flowcharts, pseudocode; Python basics: variables, data types, operators, expressions	9
II	Control flow: conditionals, loops, break/continue; functions, recursion, scope	9
III	Data structures: lists, tuples, sets, dictionaries, strings	9
IV	Files and exception handling; modules and packages; introduction to NumPy and Matplotlib	9
V	Object-oriented concepts: classes, objects, inheritance; mini-project	9
	Total contact hours (theory)	**45**

Course Outcomes (on completion, students will be able to):

(CO1) Design algorithms and flowcharts to solve simple problems.

(CO2) Write, test, and debug Python programs using appropriate constructs.

(CO3) Apply data structures and file handling to build small applications.

(CO4) Use libraries to analyse and visualise data.

Text and Reference Books: Reema Thareja, *Python Programming*, Oxford University Press; Allen Downey, *Think Python*, O'Reilly; Guido van Rossum, *Python Tutorial* (online).

[MA101] - Calculus and Linear Algebra

L-T-P-C: 3-1-0-4

Course Objectives

(a) To provide the foundations of differential and integral calculus for engineering applications.

(b) To introduce matrices, eigenvalues, and vector spaces.

Syllabus

Unit	Topics	Hours
I	Differential calculus: limits, continuity, mean value theorems, Taylor series	12
II	Functions of several variables: partial derivatives, maxima and minima, Lagrange multipliers	12
III	Integral calculus: multiple integrals, change of variables, applications	12
IV	Matrices: rank, system of equations, eigenvalues, Cayley-Hamilton theorem, diagonalisation	12
V	Vector spaces, basis, dimension, linear transformations (introduction)	12
	Total contact hours (theory + tutorial)	**60**

Course Outcomes (on completion, students will be able to):

(CO1) Apply calculus techniques to engineering problems.

(CO2) Evaluate multiple integrals and geometric quantities.

(CO3) Solve linear systems and find eigenvalues and eigenvectors.

Text and Reference Books: B. S. Grewal, *Higher Engineering Mathematics*, Khanna Publishers; Erwin Kreyszig, *Advanced Engineering Mathematics*, Wiley.

6. ELECTIVE COURSES

6.1 **Professional Electives** (examples): Artificial Intelligence, Deep Learning, Data Mining, Internet of Things, Big Data Analytics, Mobile Application Development, Blockchain Technology, Cyber Security, Embedded Systems.

6.2 **Open Electives** (examples): Entrepreneurship Development, Financial Management, Renewable Energy, Introduction to Psychology, Foreign Languages, Disaster Management.

6.3 Students shall choose electives in consultation with the faculty advisor. Electives are offered only with a minimum enrolment of [15] students.

7. LABORATORIES, PROJECTS AND INTERNSHIPS

7.1 Each laboratory course requires a record of experiments and is evaluated through continuous assessment (60 marks) and an end-semester practical examination (40 marks).

7.2 **Mini Project (Semester VI):** A team project (up to [4] students) under a faculty guide, with review presentations and a report.

7.3 **Internship:** A minimum of [4] weeks of industrial training or research internship after Semester IV or VI, with a certificate and report.

7.4 **Project Work (Semesters VII-VIII):** An individual or team project, evaluated through three reviews, a project report, and a final viva-voce examination before an external examiner.

8. MANDATORY COURSES, MOOCS AND VALUE-ADDED COURSES

8.1 Non-credit mandatory courses include Environmental Science, Constitution of India, Indian Traditional Knowledge, and Professional Ethics. A pass (satisfactory completion) is required for the award of the degree.

8.2 Credits may be earned through approved MOOCs (NPTEL / SWAYAM / Coursera), up to a maximum of [12] credits, subject to prior approval of the department and submission of the course certificate.

8.3 Value-added certificate courses are offered during semester breaks in areas such as cloud platforms, data analytics, and soft skills.

9. COURSE OUTCOME ATTAINMENT AND CURRICULUM REVISION

9.1 Attainment of Course Outcomes (COs), Programme Outcomes (POs), and Programme Specific Outcomes (PSOs) shall be assessed each semester through direct (examination results) and indirect (survey) methods.

9.2 The curriculum shall be reviewed every [two to three] years by the Board of Studies, which includes academic experts, industry representatives, and alumni.

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